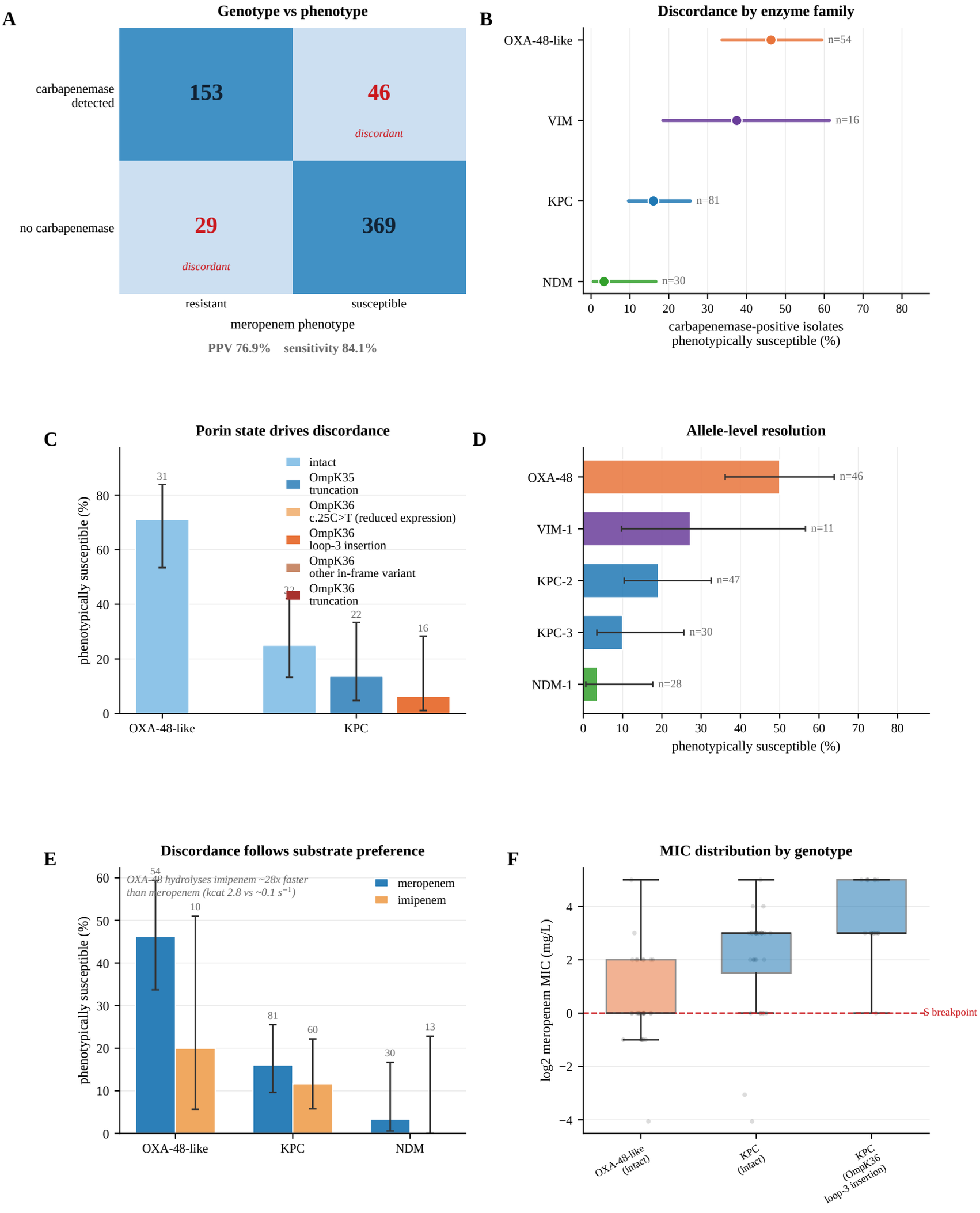
A
Genotype vs phenotype
carbapenemase detected
no carbapenemase
153
46
discordant
29
discordant
369
resistant
susceptible
meropenem phenotype
PPV 76.9%
sensitivity 84.1%
B
Discordance by enzyme family
OXA-48-like
VIM
KPC
NDM
n=54
n=16
n=81
n=30
0
10
20
30
40
50
60
70
80
carbapenemase-positive isolates phenotypically susceptible (%)
C
Porin state drives discordance
intact
OmpK35 truncation
OmpK36 c.25C>T (reduced expression)
OmpK36 loop-3 insertion
OmpK36 other in-frame variant
OmpK36 truncation
phenotypically susceptible (%)
0
20
40
60
80
31
32
22
16
OXA-48-like
KPC
D
Allele-level resolution
OXA-48
VIM-1
KPC-2
KPC-3
NDM-1
n=46
n=11
n=47
n=30
n=28
0
10
20
30
40
50
60
70
80
phenotypically susceptible (%)
E
Discordance follows substrate preference
OXA-48 hydrolyses imipenem ~28x faster than meropenem (kcat 2.8 vs ~0.1 s⁻¹)
meropenem
imipenem
phenotypically susceptible (%)
0
10
20
30
40
50
60
54
10
81
60
30
13
OXA-48-like
KPC
NDM
F
MIC distribution by genotype
log2 meropenem MIC (mg/L)
4
2
0
−2
−4
S breakpoint
OXA-48-like (intact)
KPC (intact)
KPC (OmpK36 loop-3 insertion)